DC-AC Converters (Inverters)

- Switch mode DC-AC Voltage Source Inverters (VSI)-
- Single phase Half- Bridge and
- Full-Bridge configurations-
- Sinusoidal Pulse Width Modulation (PWM) –
- Control of Fundamental output voltage-
- Harmonic spectrum-
- Bipolar and Unipolar PWM- Linear, Over Modulation and
- Square wave modes -Merits and demerits- Need for blanking time (dead-time)

INVERTERS

- DC to AC converters are known as inverters.
- An inverter changes a DC input voltage to a symmetrical AC output voltage of desired magnitude and frequency.
- The output voltage can have a fixed or variable frequency.
- Note: For AC, average value is zero (need not be sine wave)
- Applications : Variable speed ac motor drives, induction heating, standby power supply, UPS

INVERTERS

Inverters are mainly classified as:

i) Current source inverter

ii) Voltage source inverter

INVERTERS

CURRENT SOURCE INVERTER (CSI)

- This type of inverter is fed by a current source with high internal impedance.
- (A DC source in series with a large inductance will act as a current source).
- Input current can not change instantaneously.
- CSI is used in very high-power DC drives.

INVERTERS

VOLTAGE SOURCE INVERTER (VSI)

- This type of inverter is fed by a dc source of small internal impedance.

- Voltage source inverter can be classified as
  - Half-bridge inverter
  - Full-bridge inverter

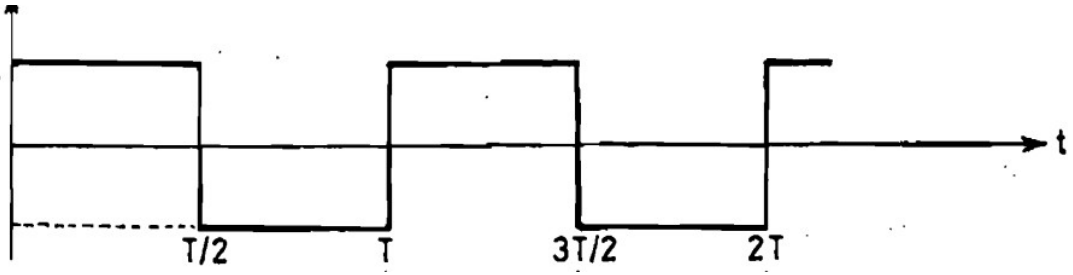
- Voltage source inverters may further be classified as
  - a) Square-wave inverter
  - b) Pulse-width modulated (PWM) inverter.

INVERTERS

a) Square-Wave Inverter

Produces a square wave ac voltage of constant value.

The output voltage can only be varied by controlling the input dc supply voltage.



INVERTERS

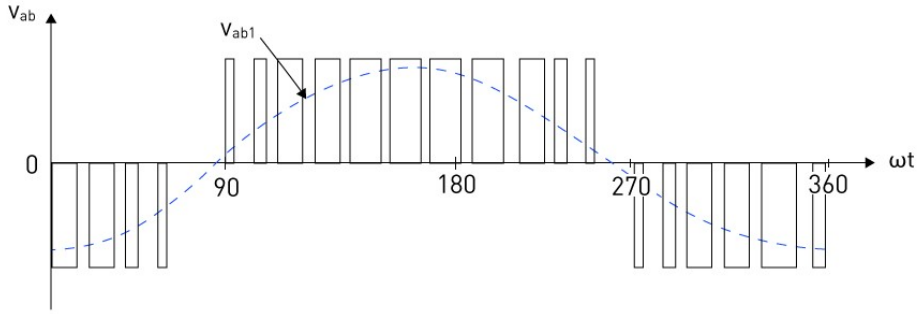
b) Pulse-Width Modulated (PWM) Inverters

The output has one or more pulses in each half-cycle.

Pulse width may be equal or different

By varying the width of these pulses, the output voltage can be controlled.

The value of the dc input voltage is essentially constant in PWM-VSI.



Single phase Inverter (DC-AC Converter)

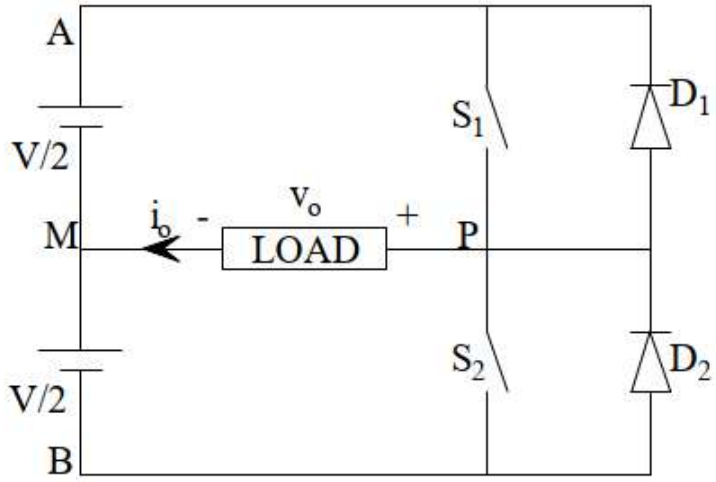
SINGLE-PHASE VOLTAGE SOURCE INVERTERS : OPERATING PRINCIPLE

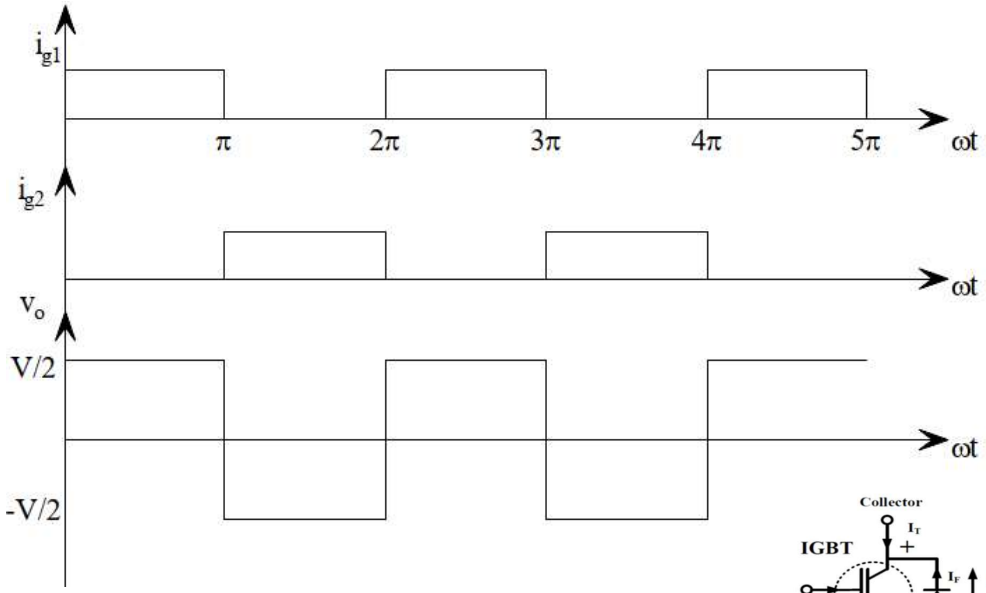
Single-phase bridge inverters are of two types,

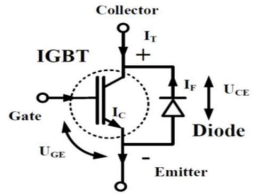
(i) Single-phase half-bridge inverters

(ii) Single-phase full-bridge inverters

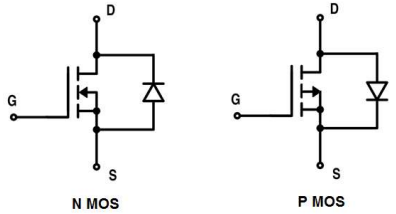
Single phase Half- Bridge Inverter





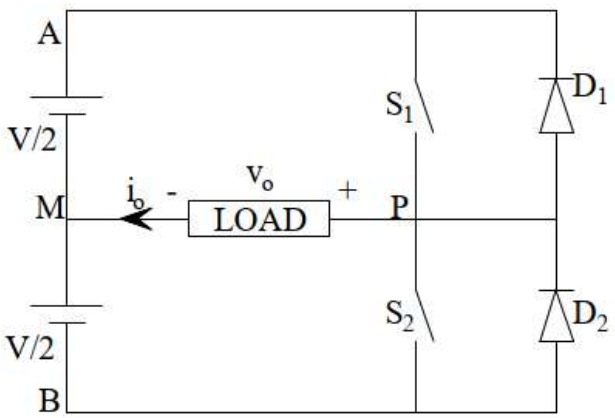


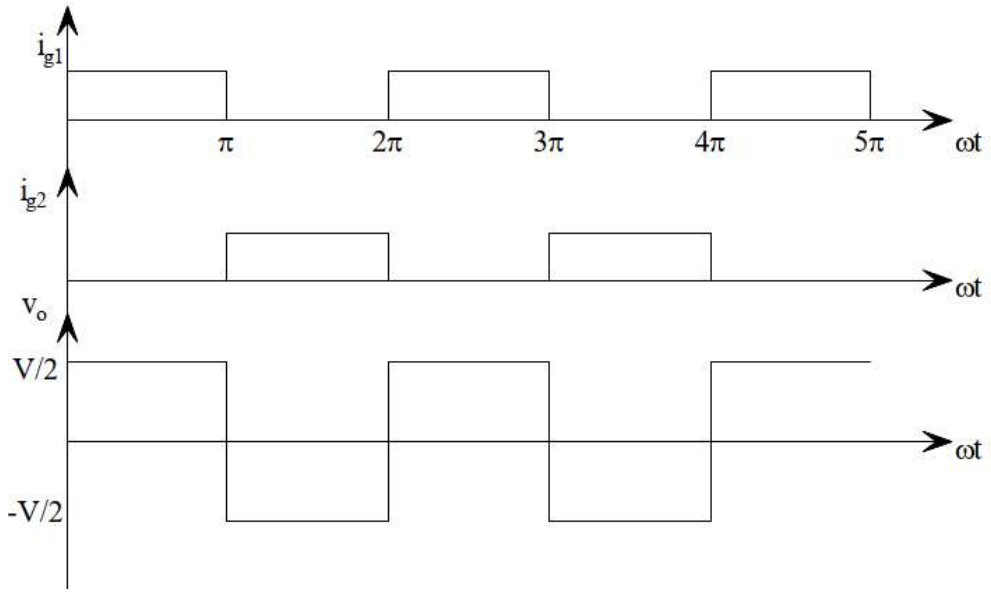
S1 and S2 are two controlled power semiconductor devices (power transistor, power MOSFET, IGBT etc) which act as switches.
Two similar sources of V/2



HALF-BRIDGE INVERTER WITH R LOAD

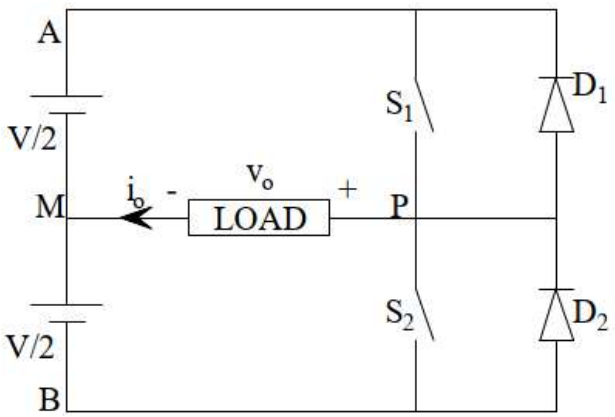
- At ωt=0, S1 is turned ON by applying a base or gate current signal.
- When S1 is conducting, load terminal P is connected to positive terminal A of DC source.
- Load voltage is V/2.

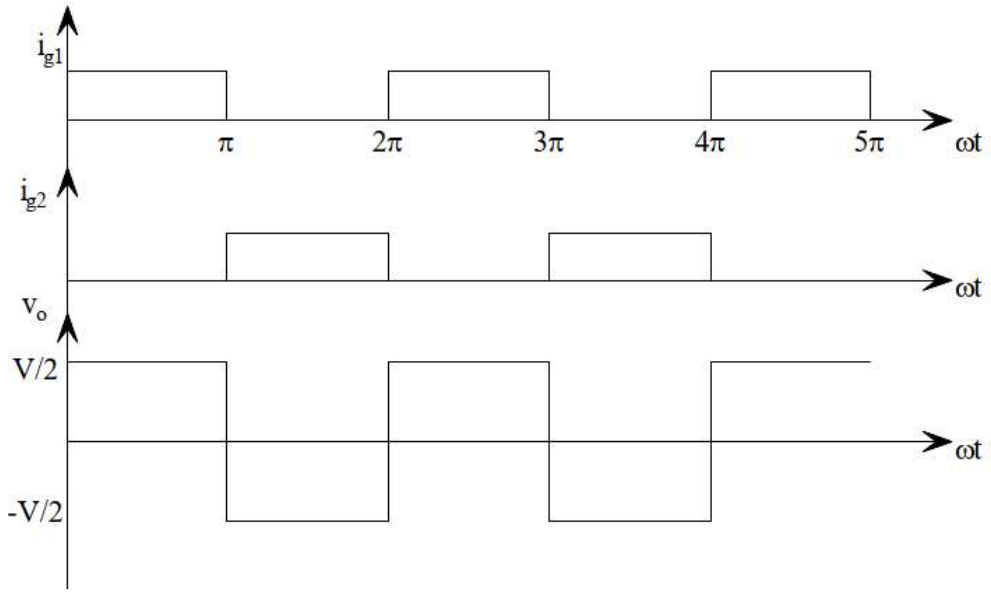




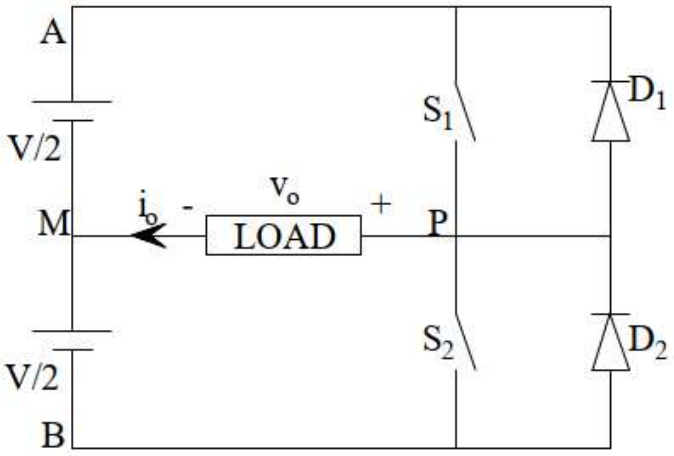
HALF-BRIDGE INVERTER WITH R LOAD

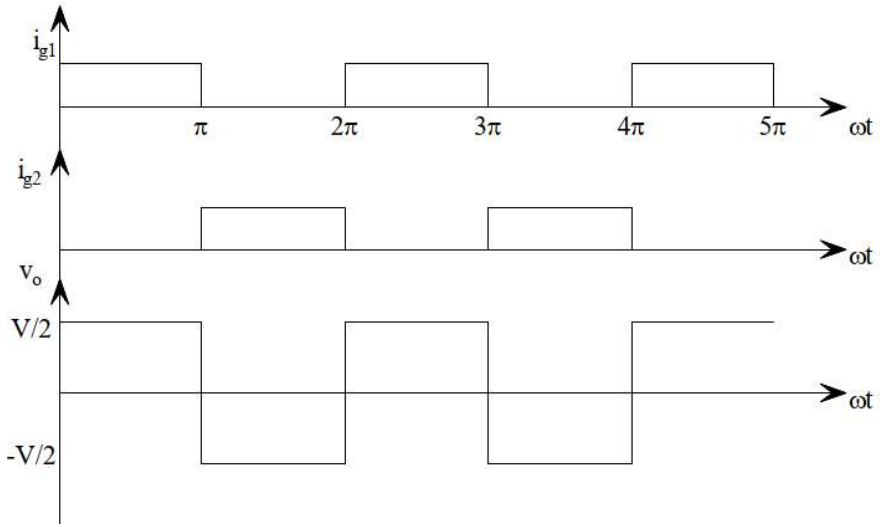
- At ωt=180°(π), base or gate signal is removed from S1 and it is turned OFF.
- At the same time, the gate signal is given to S2 and it is turned ON.
- When S2 is conducting, load terminal P is connected to negative terminal B of DC source.
- Load voltage is –V/2.





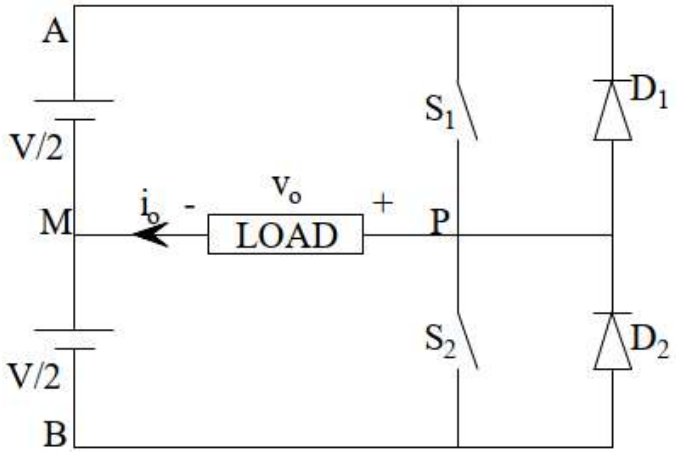
- By closing S1 and S2 alternately for half of time periods, a square-wave ac voltage is obtained at the output.
- Since conduction period of any switch is equal to duration of its base/gate current, frequency of the load voltage is controlled by control signal of the inverter.

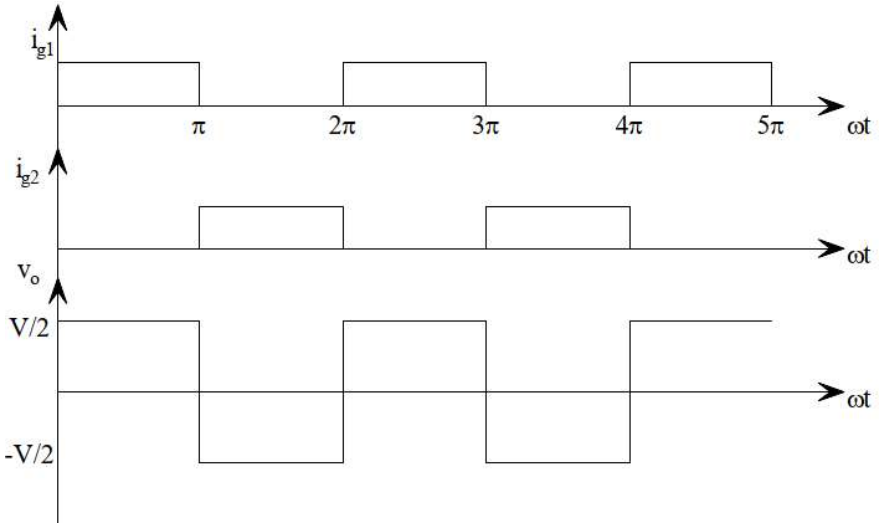




- For a pure resistive load, i_o and V_o will be in phase and of same wave shape.

- For pure resistive load, the diodes D1 and D2 do not play any role.

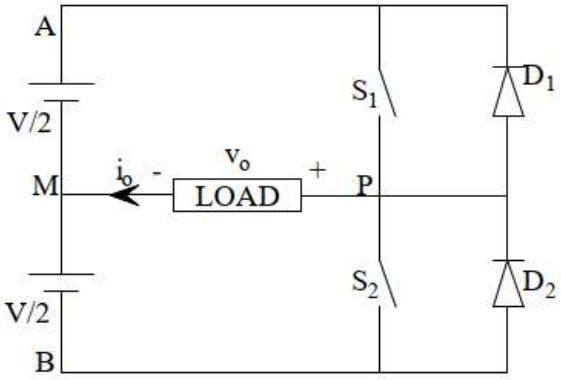


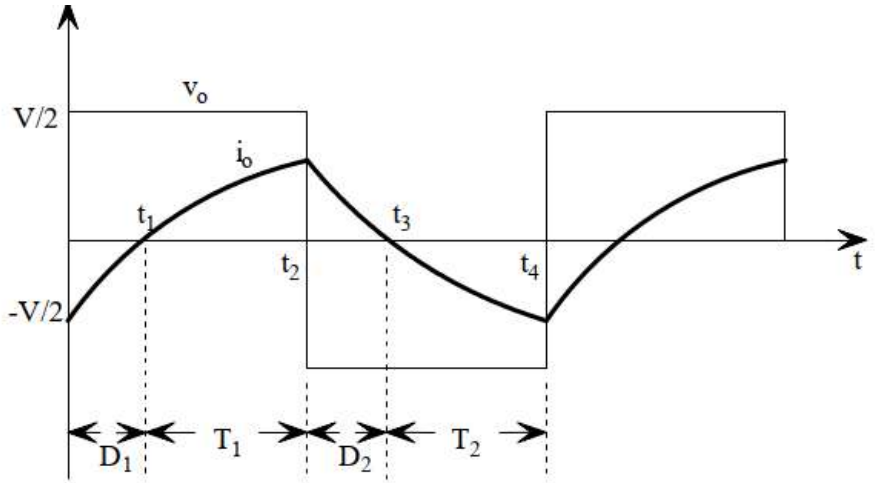


HALF-BRIDGE INVERTER WITH RL LOAD

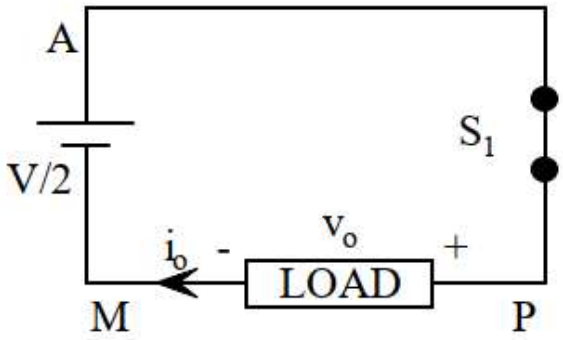
- Due to presence of inductance, load current can not be reversed instantaneously when polarity of load voltage is reversed.

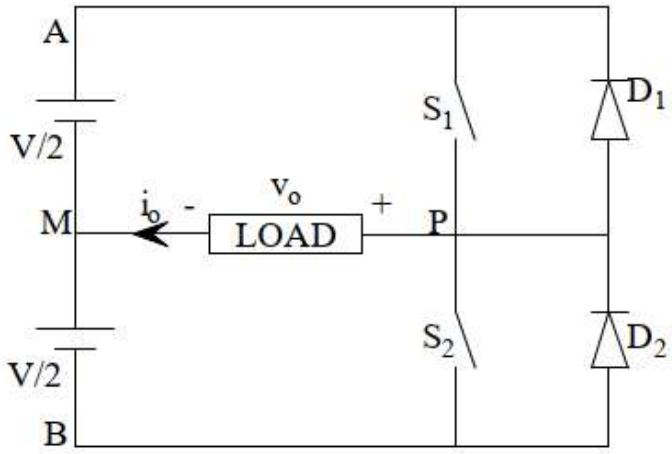
- Output voltage is similar to that with R-load.

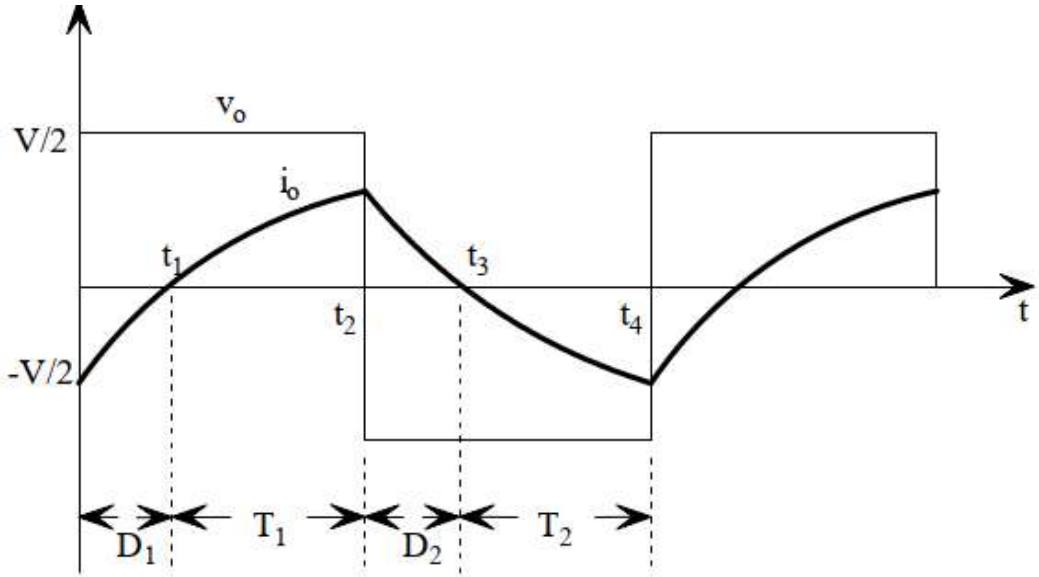




Mode 1 ($t_1 < t < t_2$)
At t_1, S_1 is turned ON.
$v_o = V/2$; load current i_o increases from zero.
At $t=t_2$, S_1 is turned OFF. Current reaches peak; energy is stored in the inductance since both v_o and i_o are positive.

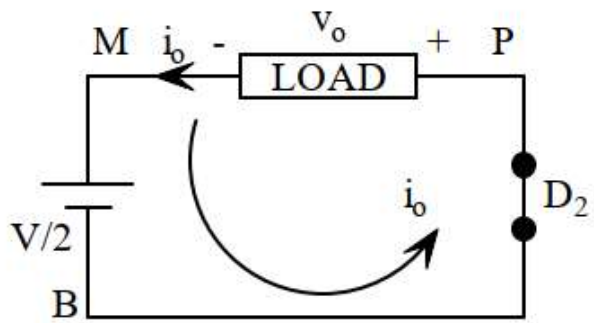


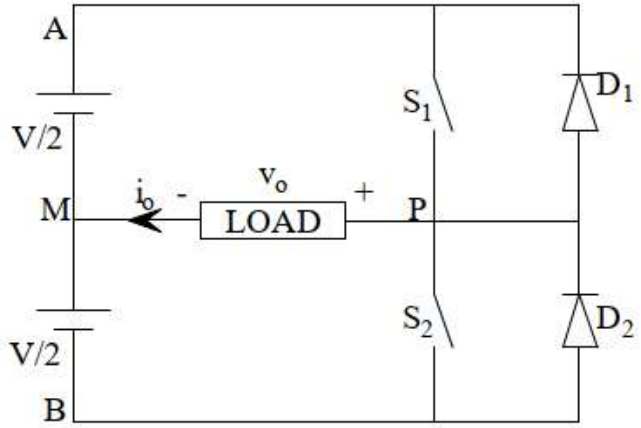


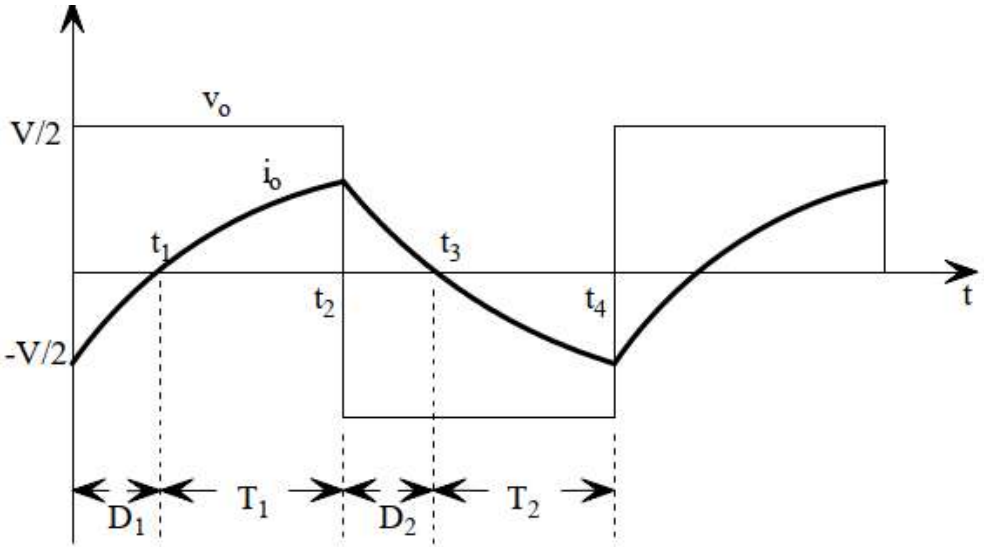


Mode 2 ($t_2 < t < t_3$)

At t_2, even though S_1 is turned OFF, load current will continue to flow in the same direction through D_2 due to stored energy in the inductance. $v_o = -V/2$; i_o decreases; stored energy is fed back to lower half of the source.

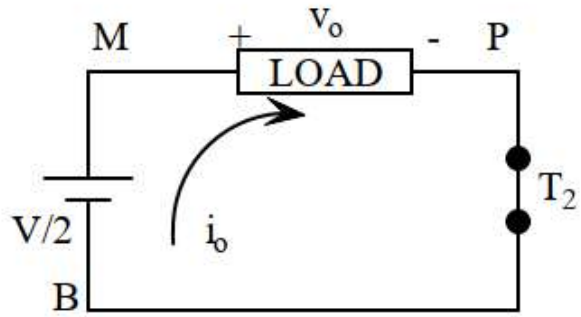


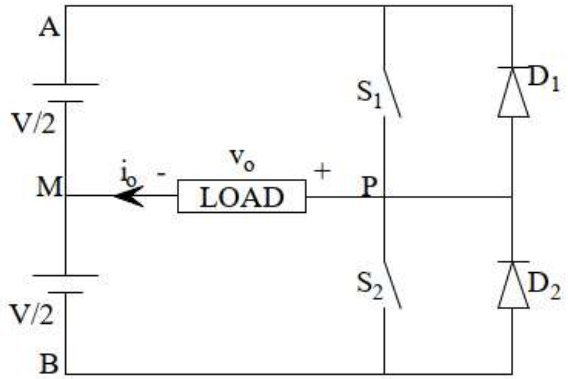


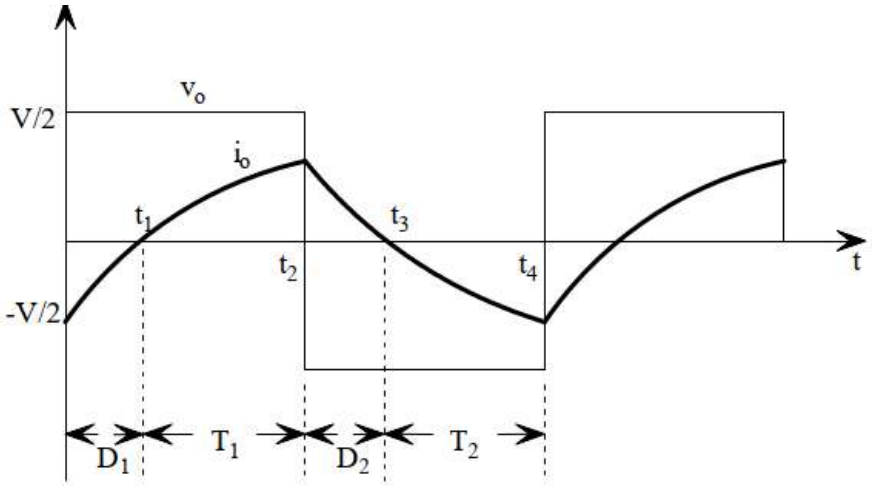


Mode 3 ($t_3 < t < t_4$)

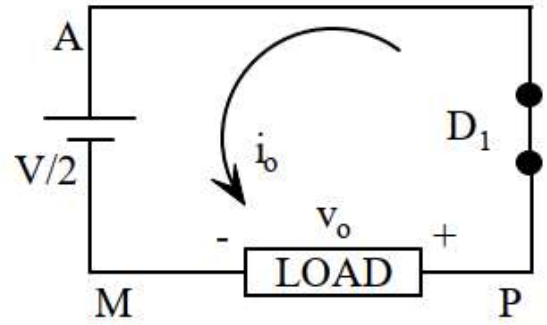
At t_3, i_o=0; S_2 is turned ON, v_o = -V/2; load current is negative.
At t_4, load current reaches peak.

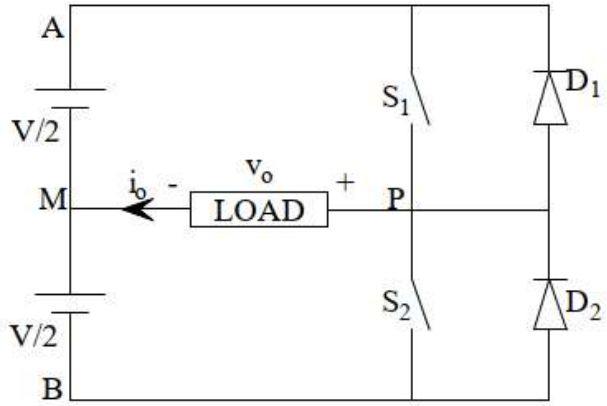


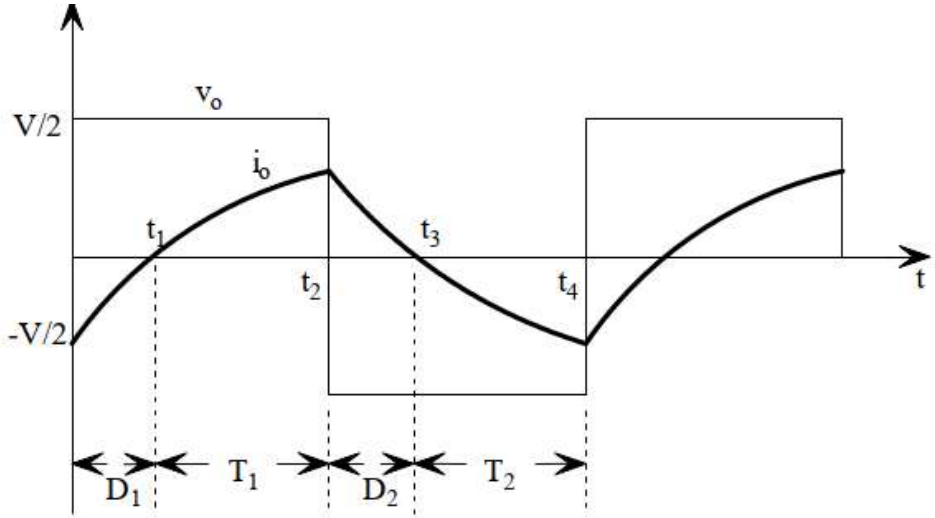




Mode 4 ($t_4 < t < t_5$) or ($0 < t < t_1$)
At t_4, i_o=0; S_2 is turned OFF, load current continues to flow through D_1. Load voltage changes polarity, $v_o = V/2$.
At t_5, i_o=0; S_1 is turned ON.







Disadvantages of half-bridge inverters are

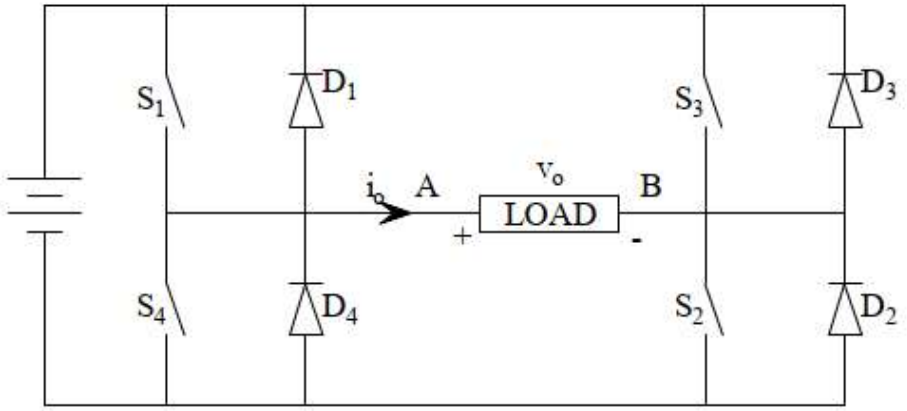
i) It requires 3-wire dc supply – dual supply (or two separate voltage sources)

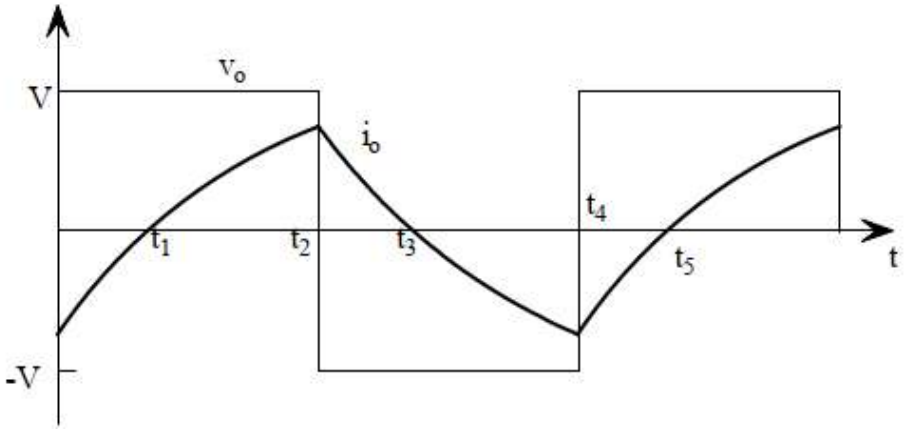
ii) Voltage across OFF state device is V which is twice the load voltage.

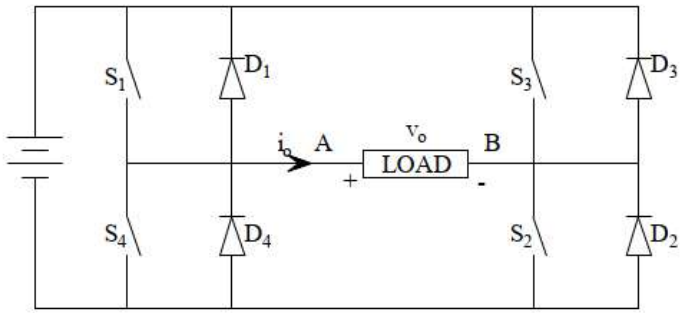
SINGLE PHASE FULL BRIDGE INVERTER

- Converts a DC power into an AC output by using four switches in an "H" configuration, like an H-bridge.
- It generates a square wave AC output by switching two pairs of switches on and off in a specific sequence
- Results in a positive voltage (+Vdc) when one pair is ON and a negative voltage (-Vdc) when the other pair is ON.

SINGLE PHASE FULL BRIDGE INVERTER with RL load



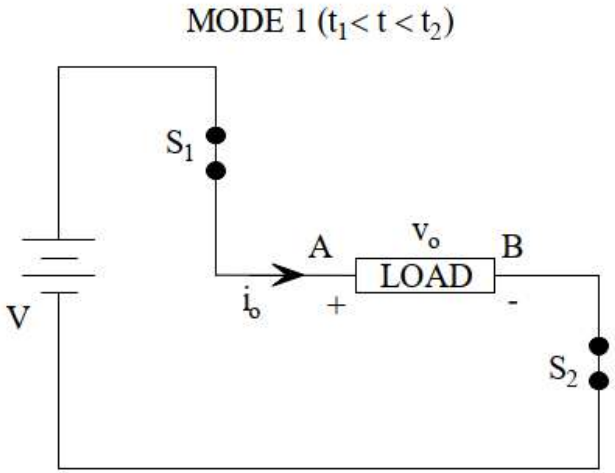


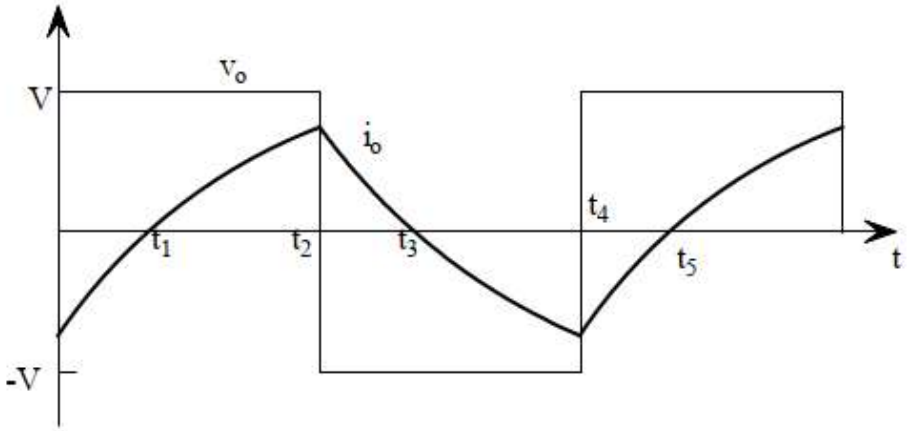


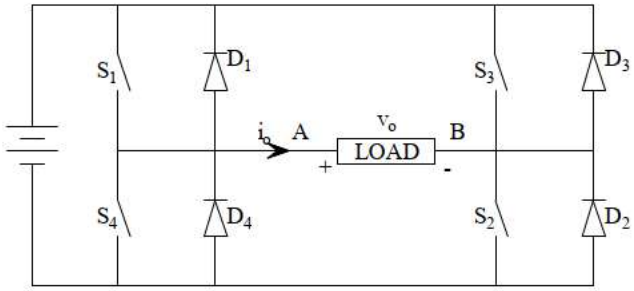
- RL load

Mode 1

At $t=t_1$; S_1 & S_2 are turned ON. $v_o = +V$; i_o starts increasing from zero. i_o reaches maximum at t_2.

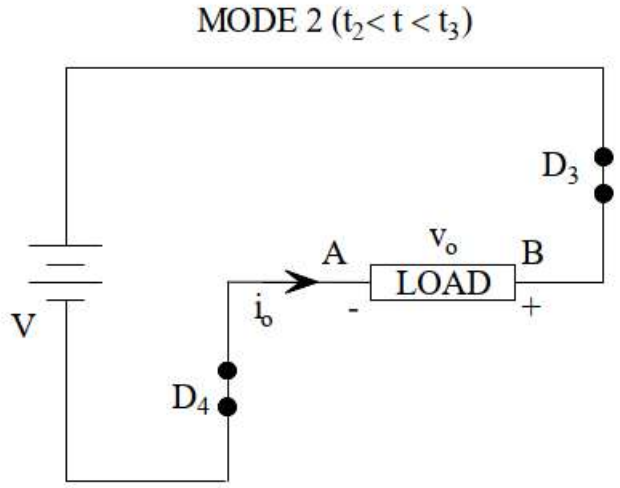


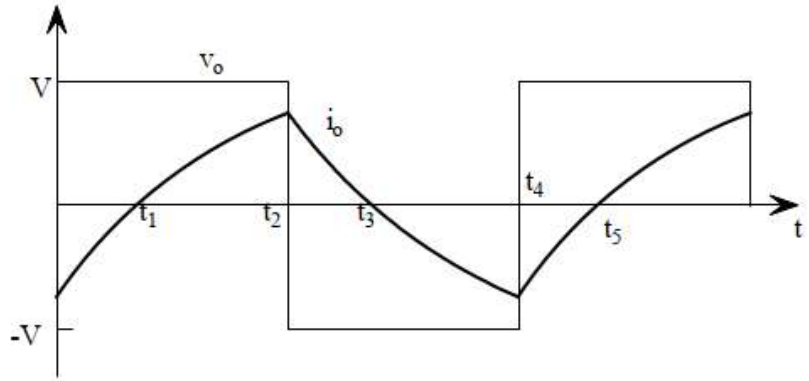


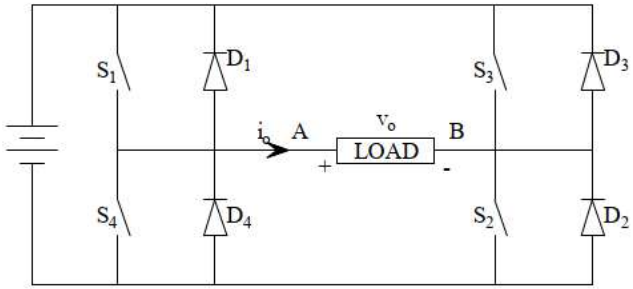


Mode 2

At $t=t_2$; S_1 & S_2 are turned OFF. i_o starts decreasing. Stored energy in the inductance is fed back to source through D_3 and D_4. $v_o = -V$.

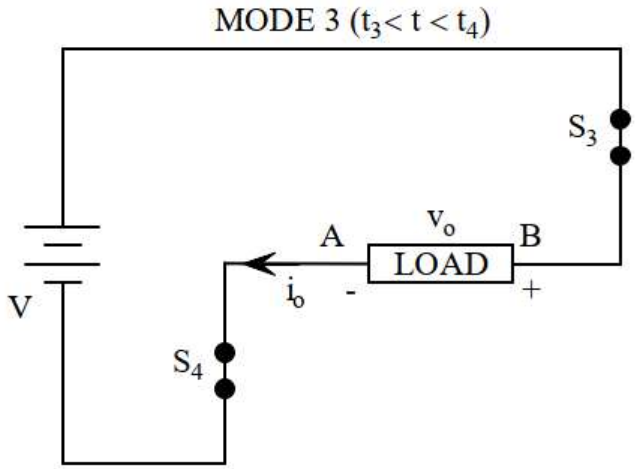


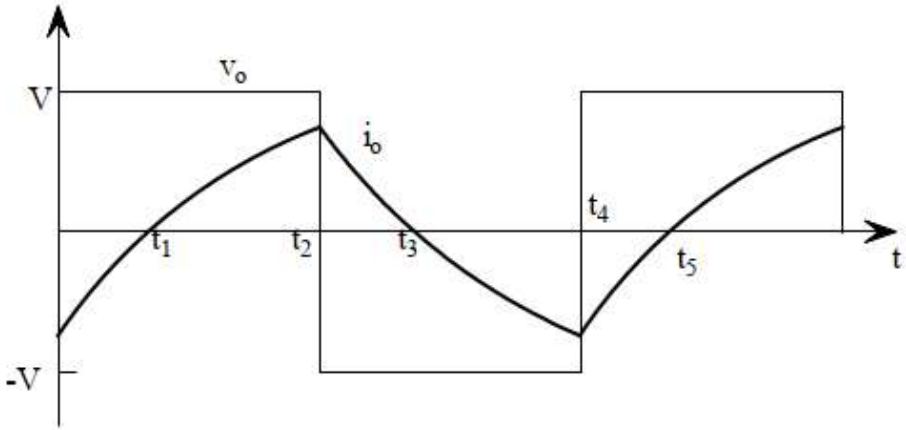


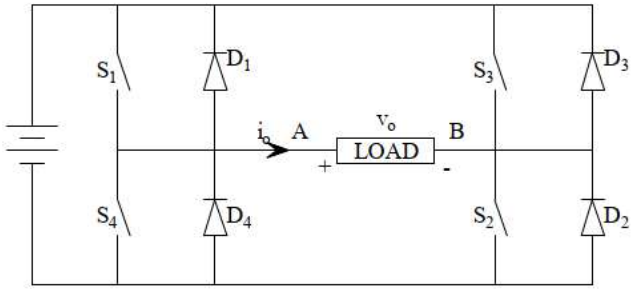


Mode 3

At t=t_3; load current becomes zero; S_3 & S_4 are turned ON; i_o flows in negative direction. v_o = -V;

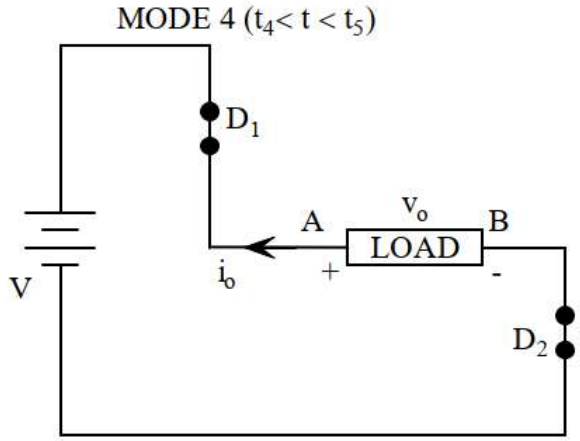


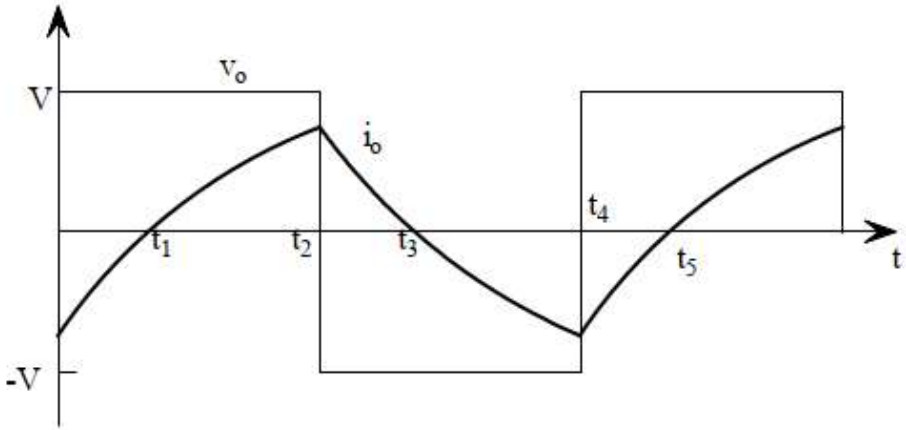




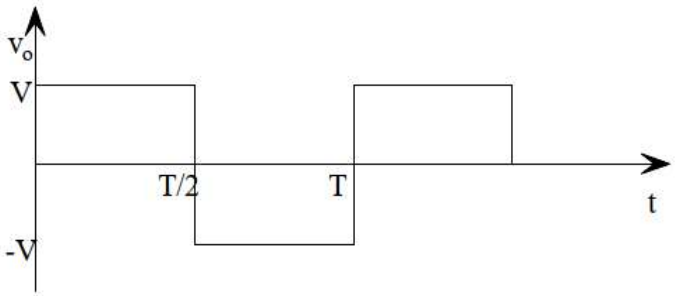
Mode 4

At t=t_4; load current becomes maximum in reverse direction; S_3 & S_4 are turned OFF; i_o continues to flows through D_1 & D_2 due to stored energy. $v_o = +V$.





ANALYSIS OF 1-PHASE FULL-BRIDGE INVERTER

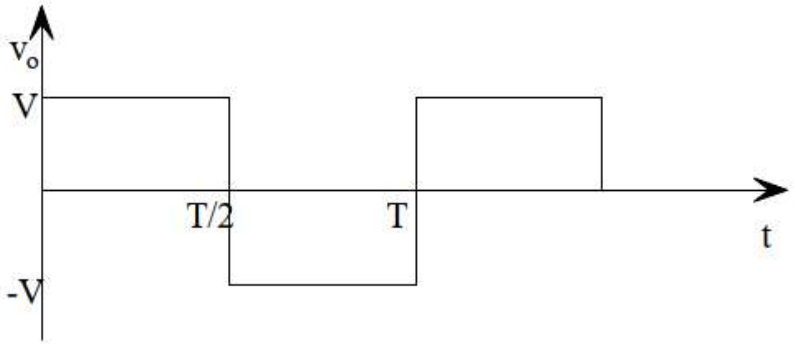


The square wave has half-wave symmetry and odd symmetry.

$$b_n = \frac{4}{2\pi}\int_0^{\pi} V \sin n\omega t d\omega t = \frac{4V}{2\pi n}[-\cos n\omega t]_0^{\pi} = \frac{4V}{n\pi} \quad \text{where n = 1,3,5,.....}$$

$$v_o(t) = \sum_{n=1,3,5,..}^{\infty} \frac{4V}{n\pi}\sin n\omega t$$

$$v_1(t) = \frac{4V}{\pi}\sin \omega t$$

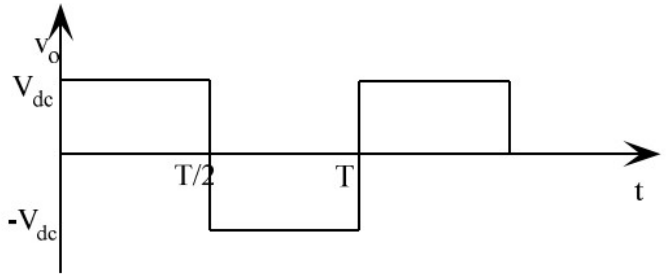


RMS value of fundamental output voltage, $V_1 = \frac{2\sqrt{2}V}{\pi} = 0.9V$

RMS value of output voltage, $V_o = \sqrt{\frac{1}{\pi}\int_0^{\pi} V^2 d\omega t} = V$

Total Harmonic Distortion, $THD = \frac{\sqrt{V_o^{\,2} - V_1^{\,2}}}{V_1} = 0.4843$

- SQUARE WAVE SWITCHING SCHEME



Fundamental component of output voltage, $v_{on}(t) = \frac{4V_{dc}}{n\pi}\sin n\omega t$

Peak value of the fundamental output voltage, $\hat{V}_{o1} = 1.273V_{dc}$

Amplitude of harmonics

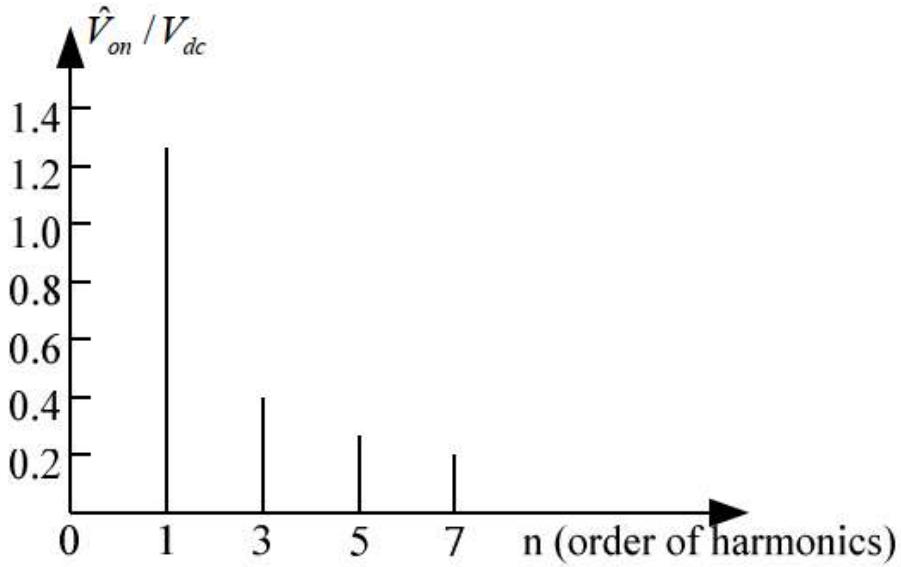
SQUARE WAVE SWITCHING SCHEME

$$\hat{V}_{Aon} = \frac{4V_{dc}}{n\pi} = \frac{1.273}{n}V_{dc}$$

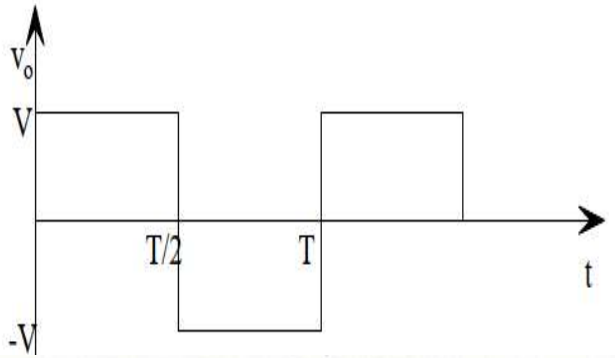
$$\hat{V}_{o3} = \frac{4V_{dc}}{3\pi} = 0.424V_{dc}$$

$$\hat{V}_{o5} = \frac{4V_{dc}}{5\pi} = 0.2546V_{dc}$$

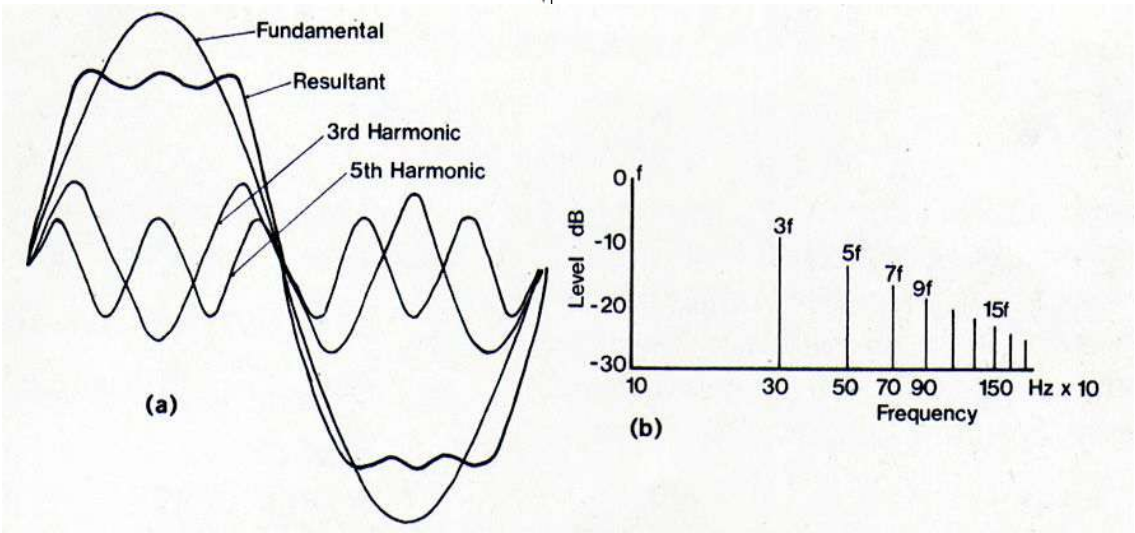
$$\hat{V}_{o7} = \frac{4V_{dc}}{7\pi} = 0.182V_{dc}$$



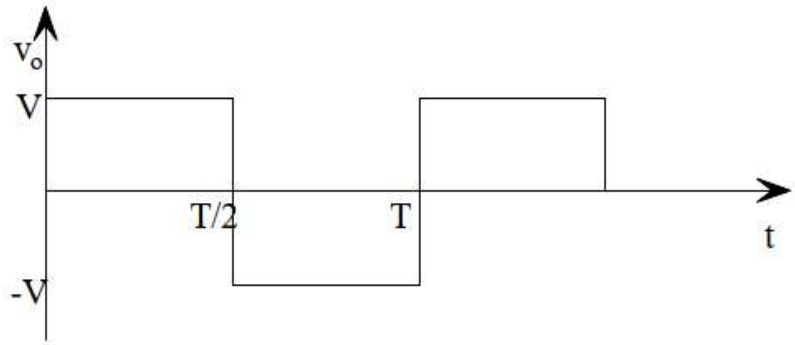
SQUARE WAVE SWITCHING SCHEME



$$v_o(t) = \sum_{n=1,3,5,..}^{\infty} \frac{4V}{n\pi} \sin n\omega t$$



Note : RMS value of fundamental, 3rd, 5th , 7th and 9th harmonics output voltages are respectively V_1=0.9V , V_3=0.9V, V_1=0.3V, V_5=0.18V, V_7=0.13V, V_9=0.1V.



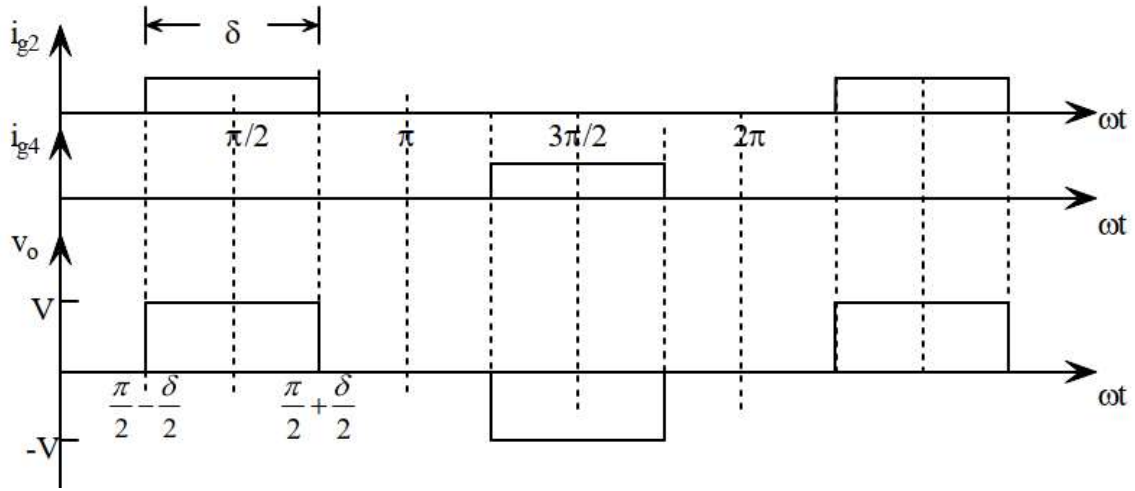
SQUARE WAVE SWITCHING SCHEME

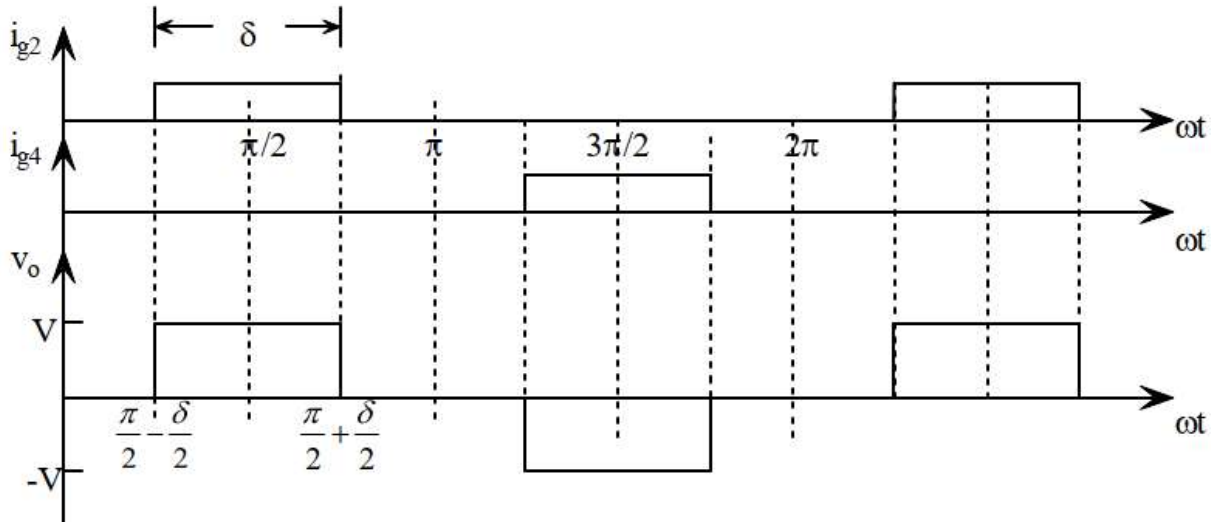
- Output voltage is independent of m_a in the square wave region.
- Output voltage can be controlled only by varying the dc input voltage Vdc.

SINGLE PULSE WIDTH MODULATION (PWM)

There is only one pulse per half-cycle and the width of the pulse is varied to control the inverter output voltage.

RMS output voltage, $V_o = \sqrt{\frac{2}{2\pi}\int_{\frac{\pi-\delta}{2}}^{\frac{\pi+\delta}{2}} V_s^2 \, d\omega t} = V_s\sqrt{\frac{\delta}{\pi}}$





By fourier analysis,

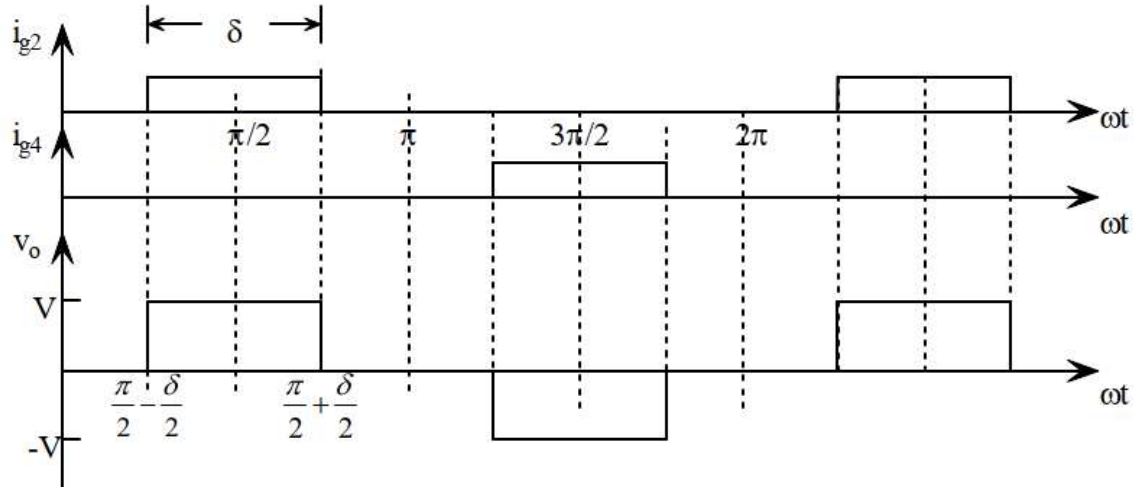
Output voltage wave has quarter-wave symmetry. [Half-wave symmetry (n is odd) & odd symmetry (a_o and a_n=0)].

By fourier analysis,

Output voltage wave has half-wave symmetry (n is odd) and odd symmetry (a_o and a_n=0)].

$$b_n = \frac{4}{2\pi}\int_{\frac{\pi-\delta}{2}}^{\frac{\pi+\delta}{2}} V\sin n\omega t d\omega t = \frac{4V}{n\pi}\sin(\frac{n\delta}{2})$$

$$V_o = \sum_{n=1,3,5..}^{\infty} \frac{4V_{dc}}{n\pi}\sin\frac{n\delta}{2}\sin n\omega t$$



$$V_o = \sum_{n=1,3,5..}^{\infty} \frac{4V_{dc}}{n\pi} \sin\frac{n\delta}{2} \sin n\omega t$$

If δ=120°, 3rd harmonics will be absent.

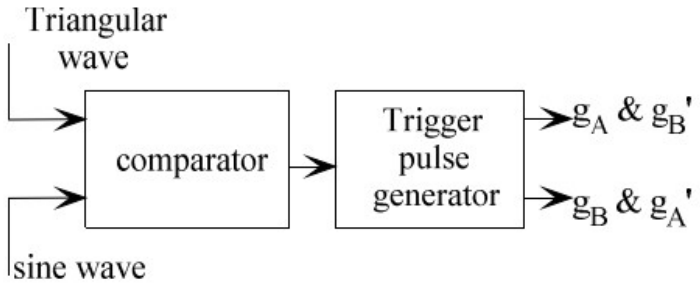
If δ=72°, 5th harmonics will be absent.

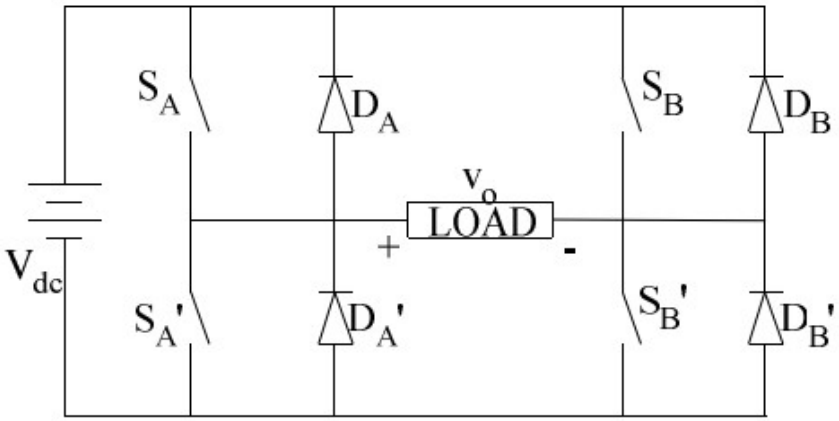
Disadvantage :- Only one harmonic can be eliminated at a time.

SINUSOIDAL PULSE WIDTH MODULATION (SPWM)

- In the sinusoidal pulse-width modulation technique,

- The amplituede of a high-frequency switching signal, called the carrier, is compared to the amplitude of a low-frequency signal, called the modulating or reference signal to generate the gate pulses



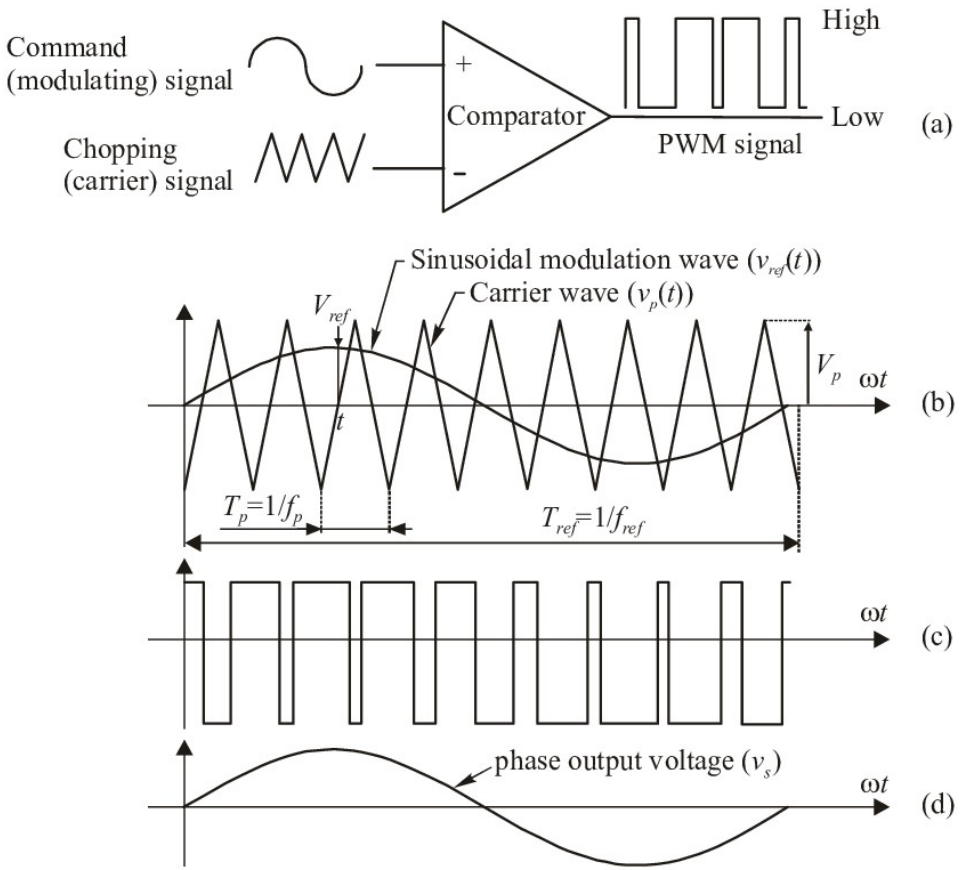


SINUSOIDAL PULSE WIDTH MODULATION (SPWM)

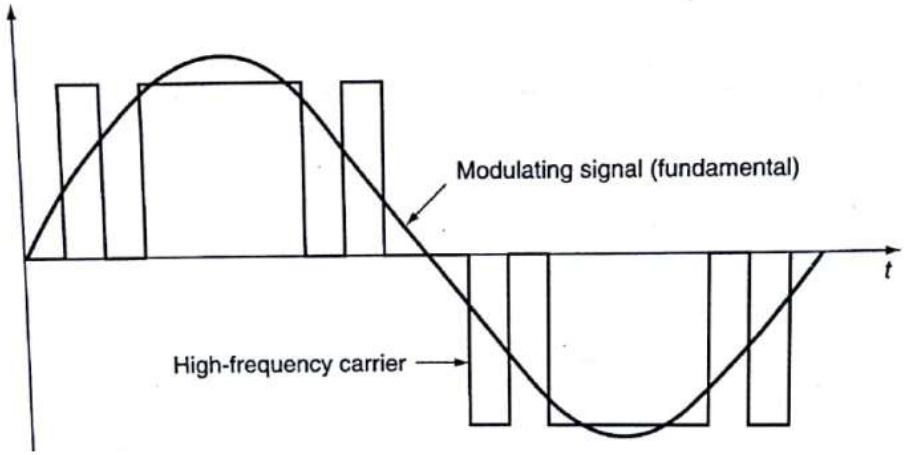
- Gating signals are generated by comparing a sinusoidal reference signal V_{sine} (of frequency f_o , same as the output frequency) with a triangular carrier wave V_{tri} of frequency f_c.

- Amplitude of carrier wave is generally kept constant,

- But varied for controlling output voltage



SPWM method

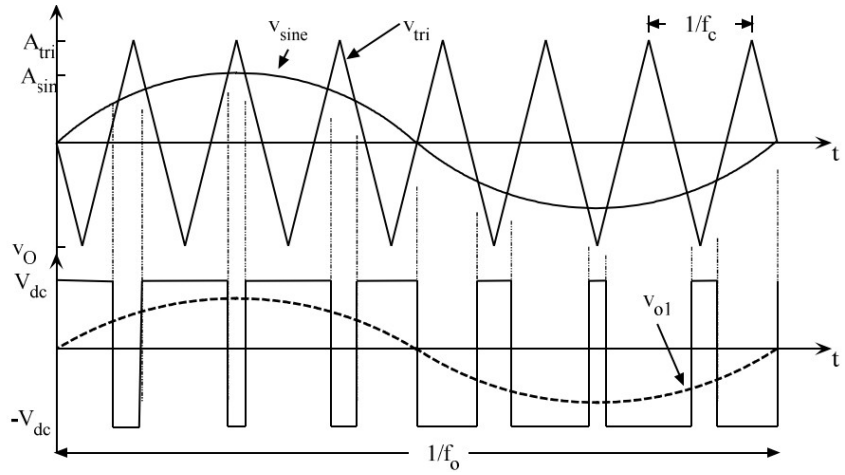


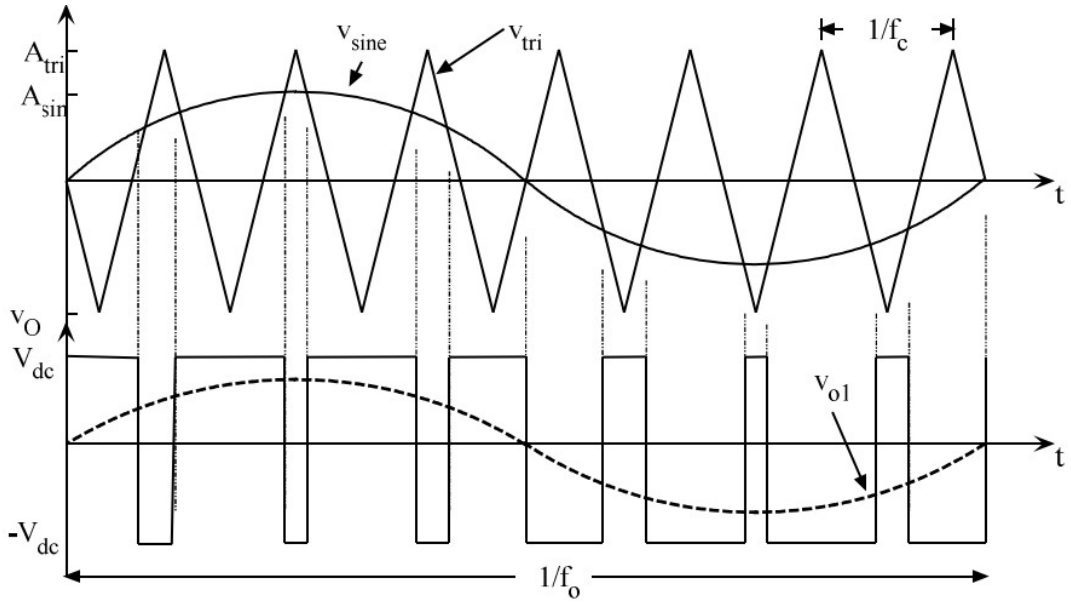
In the sinusoidal pulse-width modulation technique,

the width of a pulses at ouput of the inverter are varied in accordance to the amplitude of a sine wave, called the modulating or reference signal.

SINUSOIDAL PULSE WIDTH MODULATION (SPWM)

- Here, several pulses are used in each half-cycle of output voltage.
- Harmonic content can be reduced by this method.
- Gating signals are generated by comparing a sinusoidal reference signal with a triangular carrier wave.





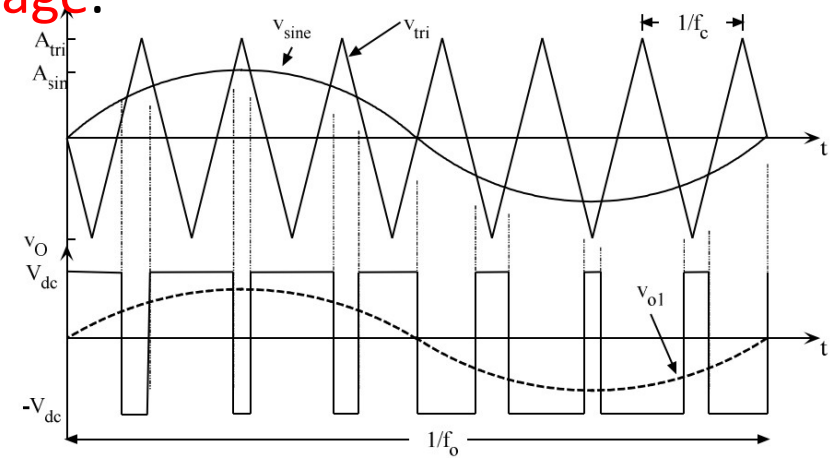
$$\text{Modulation Index} = \frac{\text{Amplitude of reference signal}}{\text{Amplitude of carrier signal}} = \frac{A_{sine}}{A_{tri}} = m_a$$

Modulation index m varies from 0 to 1.

$$\text{Frequency modulation ratio, } m_f = \frac{\text{Frequency of triangular wave}}{\text{Frequency of sine wave}} = \frac{f_c}{f_o}$$

SINUSOIDAL PULSE WIDTH MODULATION (SPWM)

- Frequency of reference signal decides the output frequency fo
- Carrier frequency fc determines the number of pulses per half-cycle.
- Modulation index controls the output voltage.

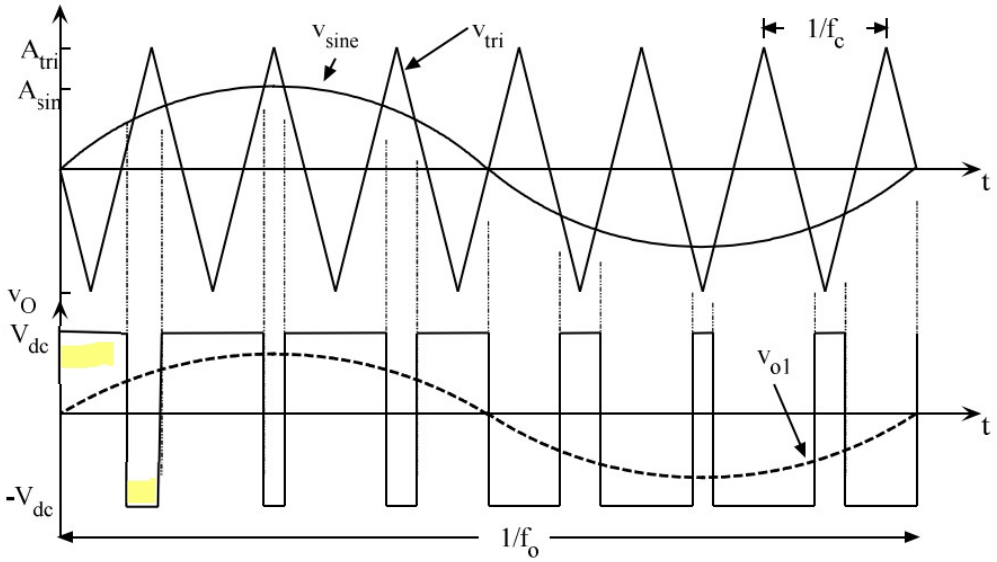


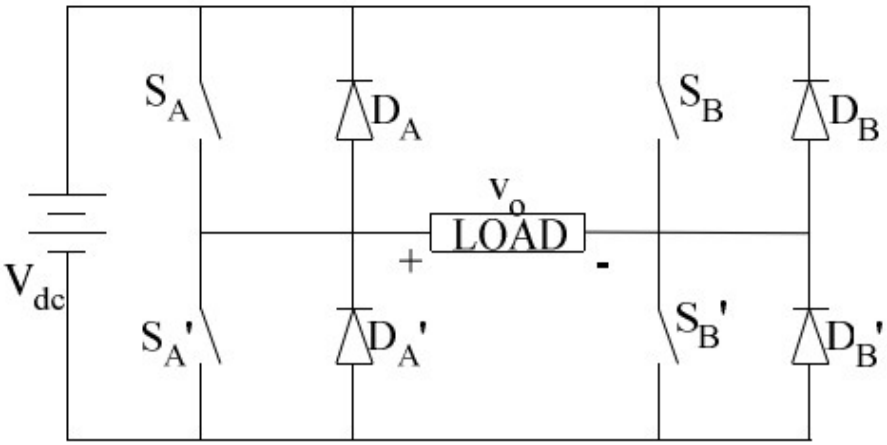
PWM WITH BIPOLAR SWITCHING

When $v_{sine} > v_{tri}$, S_A is ON (S_A' is OFF) and S_B is OFF (S_B' is ON), $v_o = +V_{dc}$

When $v_{sine} < v_{tri}$, S_A is OFF (S_A' is ON) and S_B is ON (S_B' is OFF), $v_o = -V_{dc}$

Note: Switching signals to two switches in a leg will be complimentary (When S_A is ON, S_A' will be OFF) and these two switches are never off simultaneously.





The output voltage switches between $-V_{dc}$ and $+V_{dc}$ voltage levels. Hence, the name PWM with bipolar voltage switching.

Peak of the fundamental output voltage, $\hat{V}_{o1} = m_a V_{dc}$ where $m_a \leq 1$

RMS value of the fundamental output voltage, $V_{o1} = \frac{m_a V_{dc}}{\sqrt{2}} = 0.707 m_a V_{dc}$

Peak (RMS) value of the fundamental component of the output voltage varies linearly with modulation index m_a.

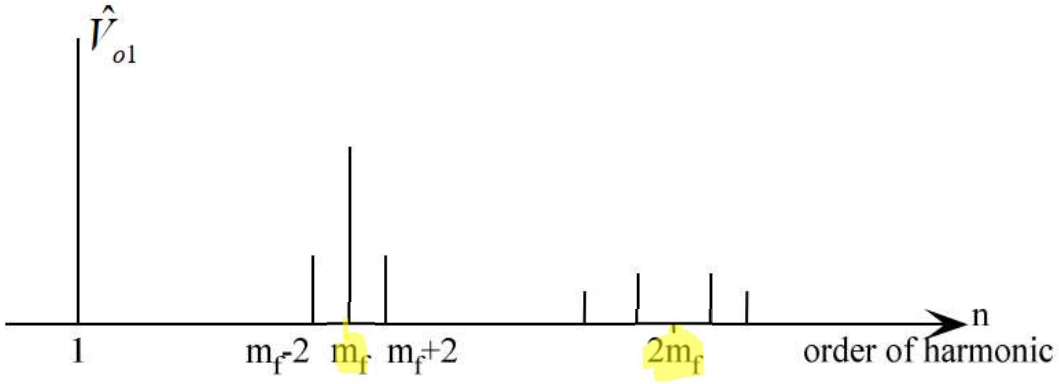
(Note : Peak value of the fundamental output voltage with square wave output,

$\hat{V}_{o1} = \frac{4V_{dc}}{\pi} = 1.273 V_{dc}$)

When $m_a > 1$, square wave operation,

$V_{dc} < \hat{V}_{o1} < \frac{4V_{dc}}{\pi}$

- The harmonics in the inverter output voltage waveform appear as side bands,
- Centered around the switching frequency and its multiples, that is, around harmonics mf, 2mf, 3mf, and so on.
- This general pattern holds true for all values of ma in the range 0-1.



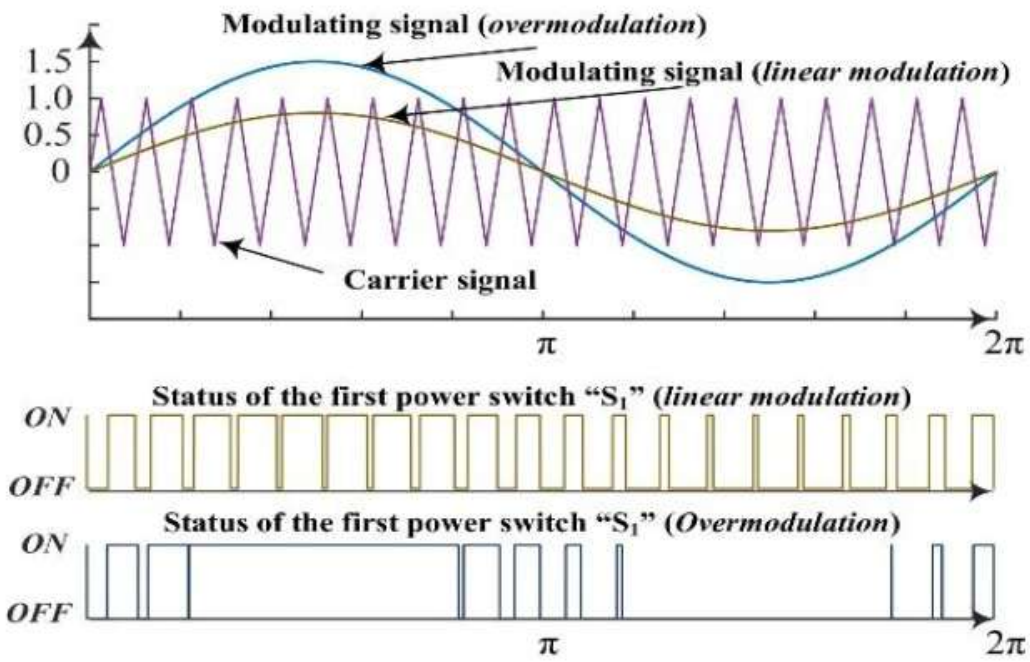
- The harmonic m_f should be an odd integer.
- Choosing m_f as an odd integer results in an odd symmetry as well as half-wave symmetry.
- Therefore, only odd harmonics are present and the even harmonics are absent.
- Moreover, only sine terms are present.

- It is desirable to use a high switching frequency as possible for eliminating the lower order harmonics.
- Higher order harmonics can be easily filtered out using a low pass filter.
- But, as switching frequency is increased, switching losses may increase and hence inverter efficiency decreases.
- By varying modulation index m_a output voltage can be controlled.

LINEAR MODULATION & OVERMODULATION

- If ma > 1 leads to over-modulation which causes the output voltage to be a square wave (square wave operation).

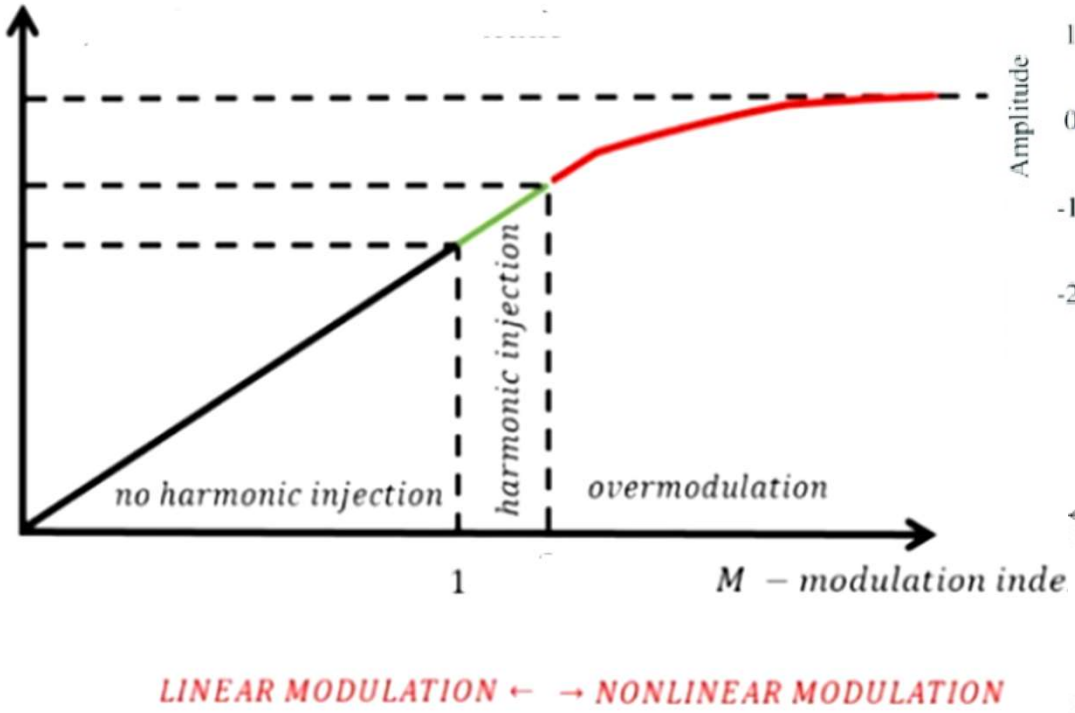
- For ma > 1 the output is less controllable as the pulse width is not controllable.

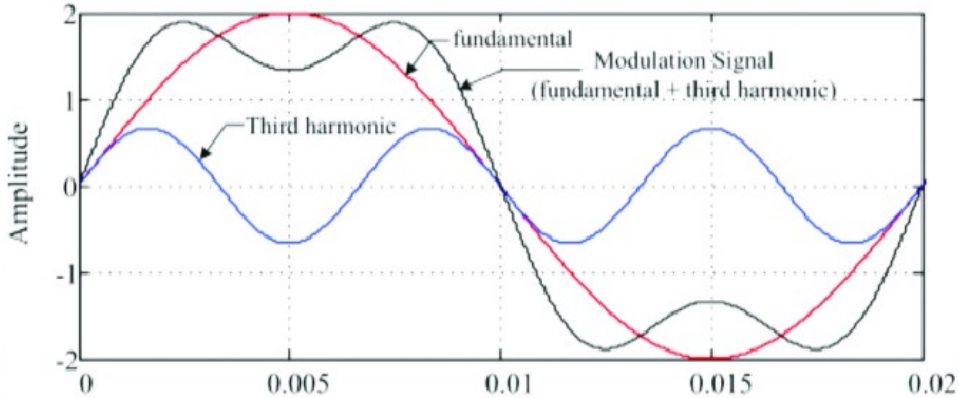


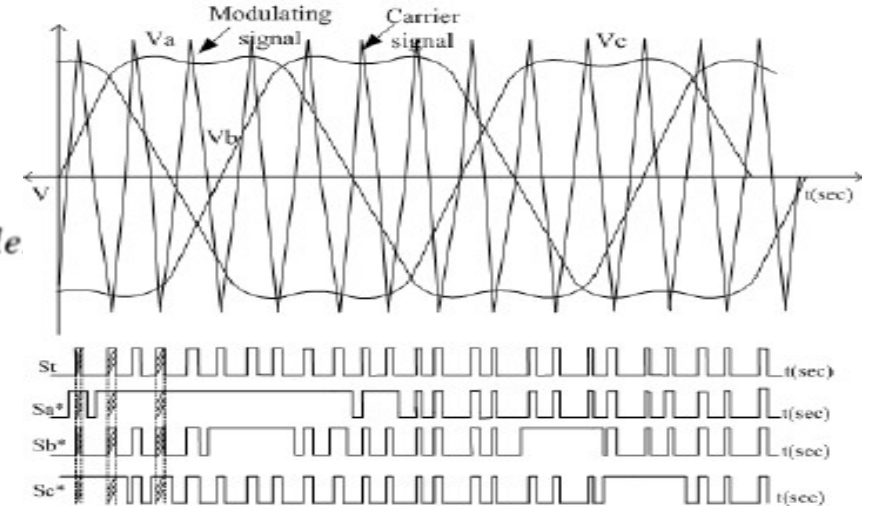
LINEAR MODULATION & OVERMODULATION

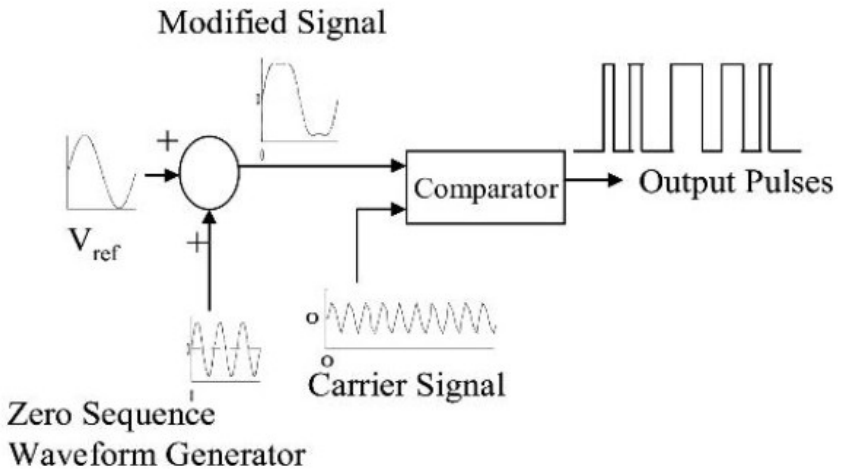
- With over-modulation, fundamental output voltage increases
- But it causes more harmonics.
- Overmodulation causes the output voltage to contain many more harmonics in the side-bands as compared with the linear range (with ma < 1).
- Hence, over-modulation is normally avoided in applications requiring low distortion.

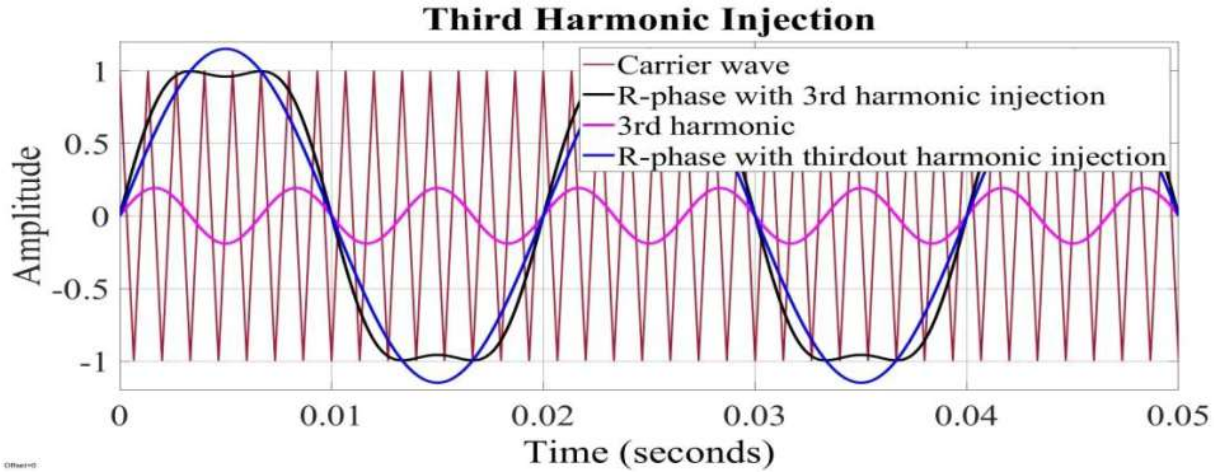
LINEAR MODULATION by Third harmonic injection








Modified Signal
+
+
Comparator
Output Pulses
V_{ref}
Carrier Signal
Zero Sequence
Waveform Generator

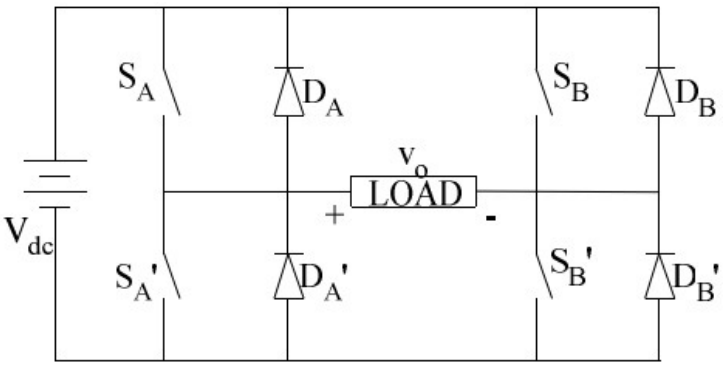

Third Harmonic Injection
Carrier wave
R-phase with 3rd harmonic injection
3rd harmonic
R-phase with thirdout harmonic injection
Amplitude
1
0.5
0
-0.5
-1
0
0.01
0.02
0.03
0.04
0.05
Time (seconds)

PWM WITH UNIPOLAR SWITCHING

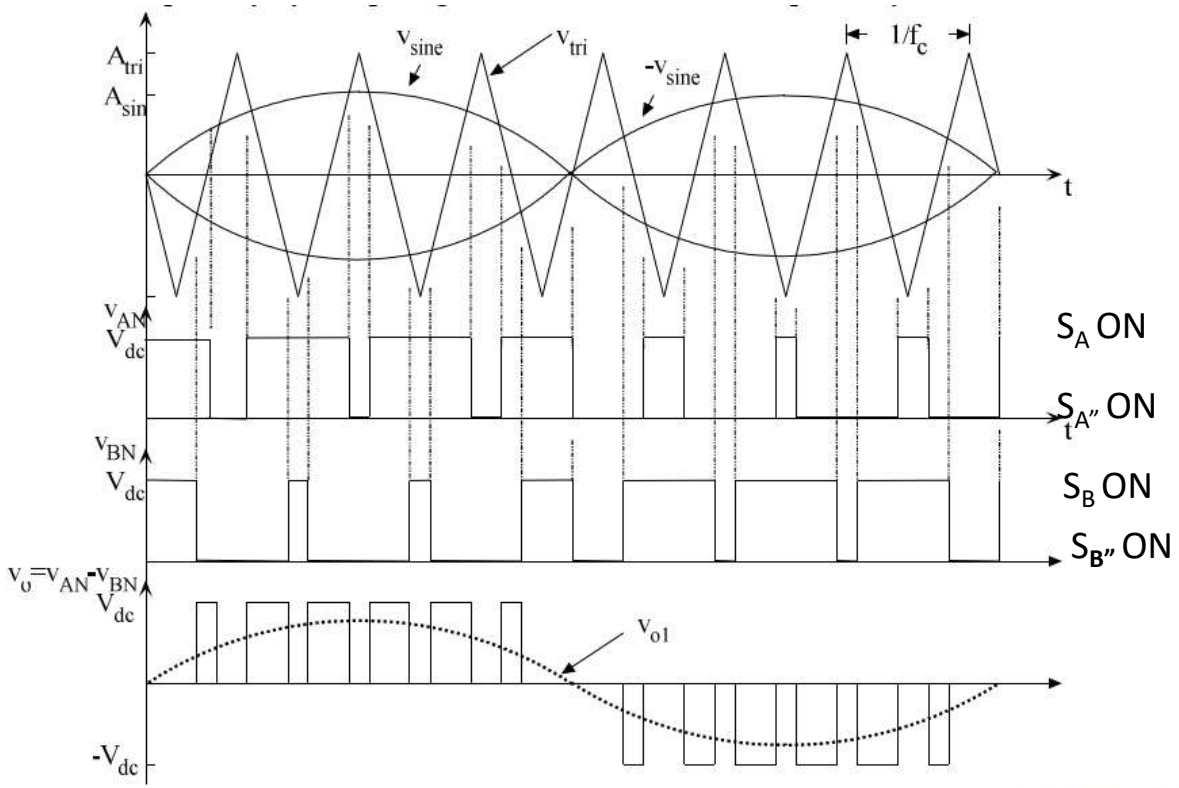
In PWM with unipolar voltage switching,

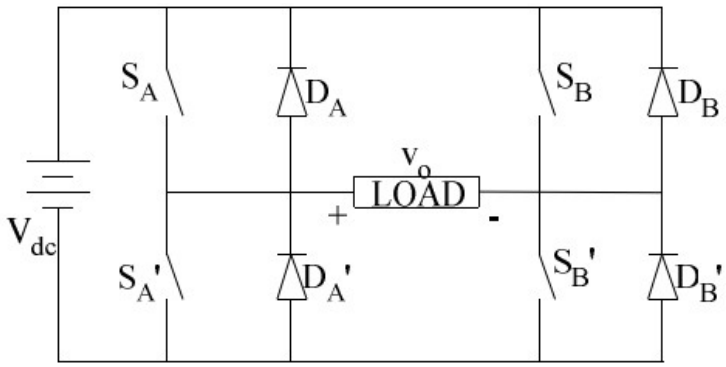
- the switches in the two legs of the full-bridge inverter is not switched simultaneously.

- Here, the legs A and B of the full-bridge inverter are controlled separately by comparing vtri with vsine and –vsine respectively.



PWM WITH UNIPOLAR SWITCHING





When $v_{sine} > v_{tri}$, S_A is ON, $v_{AN} = V_{dc}$

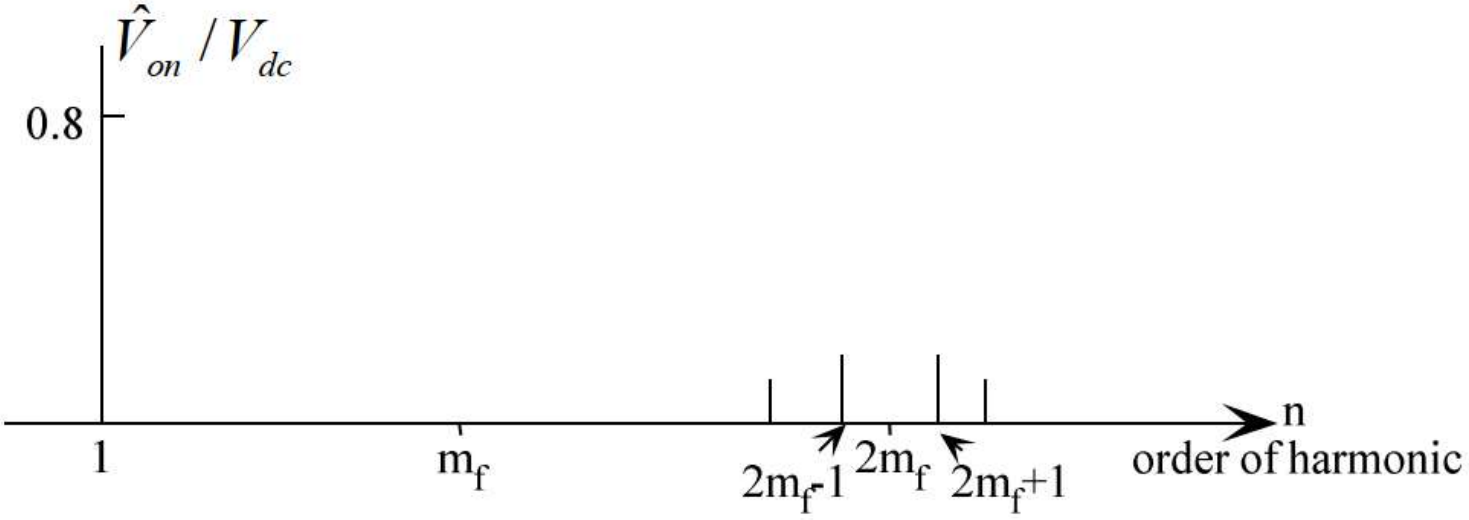
When $v_{sine} < v_{tri}$, S_A' is ON, $v_{AN} = 0$

When $-v_{sine} > v_{tri}$, S_B is ON, $v_{BN} = V_{dc}$

When $-v_{sine} < v_{tri}$, S_B' is ON, $v_{BN} = 0$

Output voltage, $v_o = v_{AN} - v_{BN}$

- In this type of PWM scheme, when a switching occurs, the output voltage changes between zero and +Vdc or between zero and –Vdc voltage levels.
- Hence, this type of PWM scheme is called PWM with a unipolar voltage switching.
- By this switching scheme, the switching frequency in output voltage is "effectively" doubled.
- Lowest harmonics appear as sidebands of twice the switching frequency.
- The harmonics in the inverter output appear as sidebands around 2mf , 4mf , 6mf, …



- Note the harmonics at and around mf , 3mf , 5mf , … are absent,
- hence lower harmonic content compared to bipolar switching.

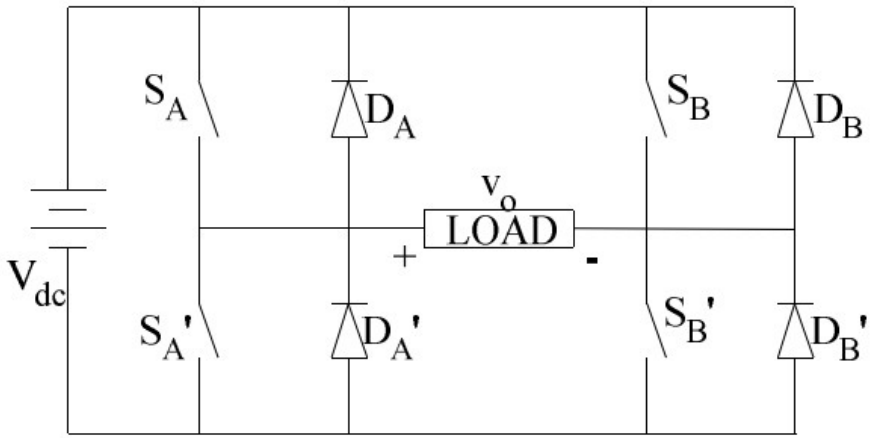
- Also, the current ripples are less with unipolar switching.

- Here, in unipolar switching scheme mf is chosen to be even whereas in bipolar switching scheme, mf is odd.

- Note also only odd harmonics are present.

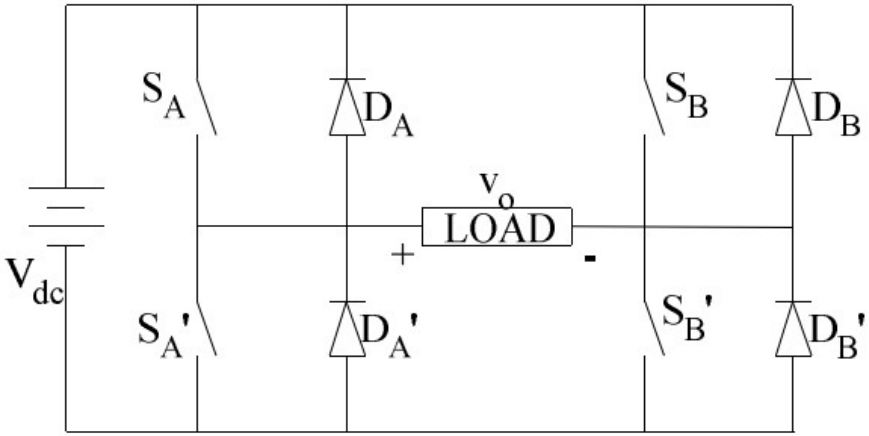
NEED FOR BLANKING TIME

- Blanking time in inverters, also known as dead time,
- It is a brief period when both switches in an inverter leg are off to prevent a short circuit.
- Its effects include improved safety by preventing shoot-through.



NEED FOR BLANKING TIME

- It can cause power quality issues.
- Decreased fundamental voltage, increased harmonic distortion, and reduced efficiency.
- The exact effect depends on the blanking time's duration, the current's direction, and the load type.



Advantages

- Shoot-through prevention:
- The primary purpose of blanking time is to prevent shoot-through,
- a condition where both the upper and lower switches in the same inverter leg conduct simultaneously, creating a short circuit across the DC link and potentially damaging the inverter.

- Diode conduction: During the blanking time, the current is carried by the freewheeling diodes in the inverter leg, which prevents the inverter from being damaged.

Disadvantages

- Harmonic distortion: Blanking time can distort the output voltage and current waveforms, leading to a decrease in the fundamental component and an increase in low-order harmonics, which can negatively impact the performance of connected loads.
- Reduced efficiency:
- Too short: A blanking time that is too short may not fully prevent shoot-through, leading to increased switching losses and reduced efficiency.
- Too long: An excessively long blanking time can lead to increased power losses, also decreasing overall efficiency.

Disadvantages

- Voltage error: The output voltage error caused by blanking time is dependent on the current flowing through the switch at the moment of switching.
- This error is more significant at low output voltages.

- Dependence on current direction: The effect of blanking time varies with the sign (direction) of the current.

- A positive current can lead to a decrease in pulse width, while a negative current can cause an increase in pulse width, creating waveform distortion.